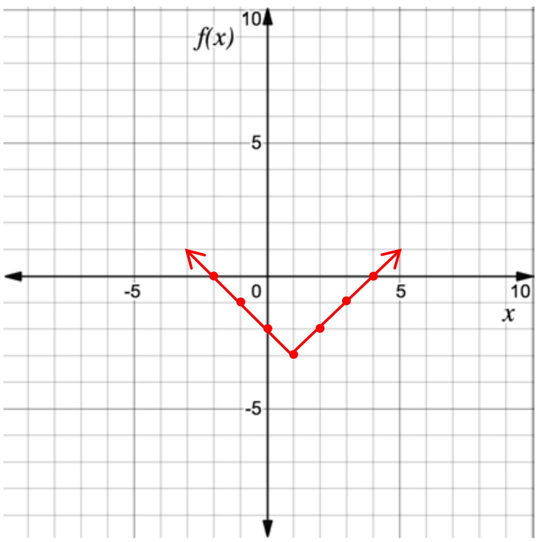
Algebra 1
Unit 10
Lesson 1 Practice

Name____________________
Date____________________
Period___________________

1. Complete the table of values for $f(x) = |x-1| - 3$.

$\boldsymbol{x}$	-2	-1	0	1	2	3	4
$\boldsymbol{f(x)}$	0	-1	-2	-3	-2	-1	0

2. Use the table of values to graph $f(x) = |x-1| - 3$ on the coordinate grid.



3. What is the vertex of $f(x) = |x-1| - 3$?
$(1, -3)$

4. What is the equation for the axis of symmetry of $f(x) = |x-1| - 3$?
$x = 1$

5. What are the x and y-intercept(s) of $f(x) = |x-1| - 3$?
x-int: $(-2,0)(4,0)$
y-int: $(0,-2)$

6. What are the domain and range of $f(x) = |x-1| - 3$?
Domain: All real numbers
Range: $y \geq -3$

7. Identify the intervals on which the function $f(x) = |x - 1| - 3$ is increasing and/or decreasing.

   Increasing: $1 < x < \infty$

   Decreasing: $-\infty < x < 1$

8. Identify the intervals on which the function $f(x) = |x - 1| - 3$ is positive and negative.

   Positive: $x < -2$ or $x > 4$

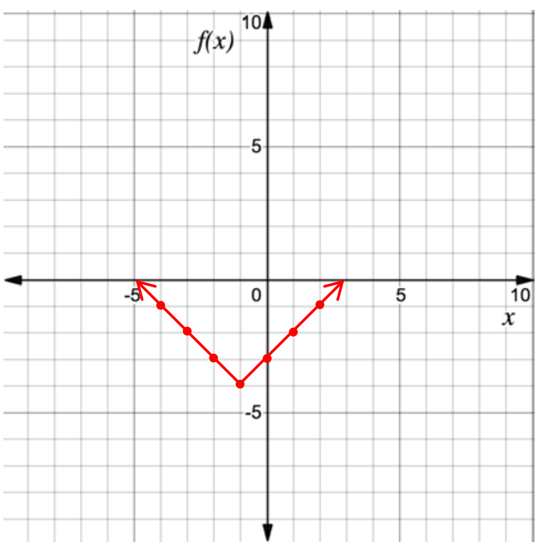
   Negative: $-2 < x < 4$

For questions 9-10, consider the absolute value function $f(x) = |x + 1| - 4$.

9. Complete the table of values for $f(x) = |x + 1| - 4$.

$\boldsymbol{x}$	-4	-3	-2	-1	0	1	2
$\boldsymbol{f(x)}$	-1	-2	-3	-4	-3	-2	-1

10. Use the table of values to graph $f(x) = |x + 1| - 4$ on the coordinate grid.



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Lesson 2 Practice Period______________________

Use the following context to answer questions 1-5.

The average mass of a baseball is $145\frac{1}{2}$ grams (g) with a tolerance of $3\frac{1}{2}$ g. If x is the mass of the baseball and V is the variation from $145\frac{1}{2}$ g., the function $V(x) = \left|x - 145\frac{1}{2}\right|$ can be used to find the amount of variation.

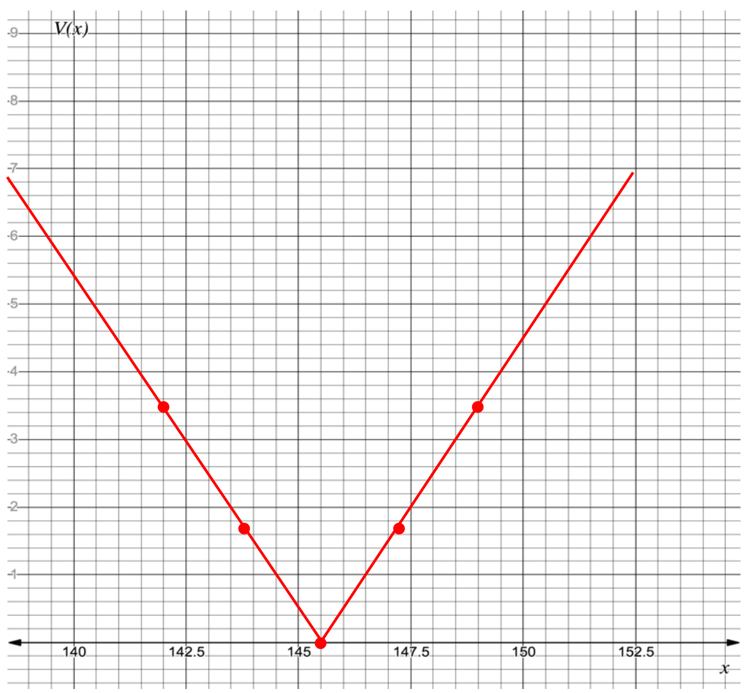
1. Determine the maximum and minimum x-values.

minimum = 142 grams maximum = 149 grams

2. Complete the table for $V(x) = \left|x - 145\frac{1}{2}\right|$. Use the minimum x-value in the first column, the maximum x-value in the last column, and add $1\frac{3}{4}$ to each x-value in between.

x	142	$143\frac{3}{4}$	$145\frac{1}{2}$	$147\frac{1}{4}$	149
y	$3\frac{1}{2}$	$1\frac{3}{4}$	0	$1\frac{3}{4}$	$3\frac{1}{2}$

3. Sketch the graph of the absolute value function



4. Interpret the vertex of $V(x) = \left|x - 145\frac{1}{2}\right|$.

The average mass of a baseball.

5. Interpret the axis of symmetry of $V(x) = \left|x - 145\frac{1}{2}\right|$.

The line that determines how far away from the average mass of a baseball, in grams, in either direction the mass varies.

Use the following context to answer questions 6-10.

The clearance below a bridge is 185.5 feet (ft) above sea level. According to the distribution of tidal phases, the water under the bridge can fluctuate up to 3 feet above or below the normal depth. If x is the amount of clearance the bridge has above the water, in ft, and W is the variation in water depth away from sea level, in ft, the equation to find the amount of variation is $W(x) = |x - 185.5|$.

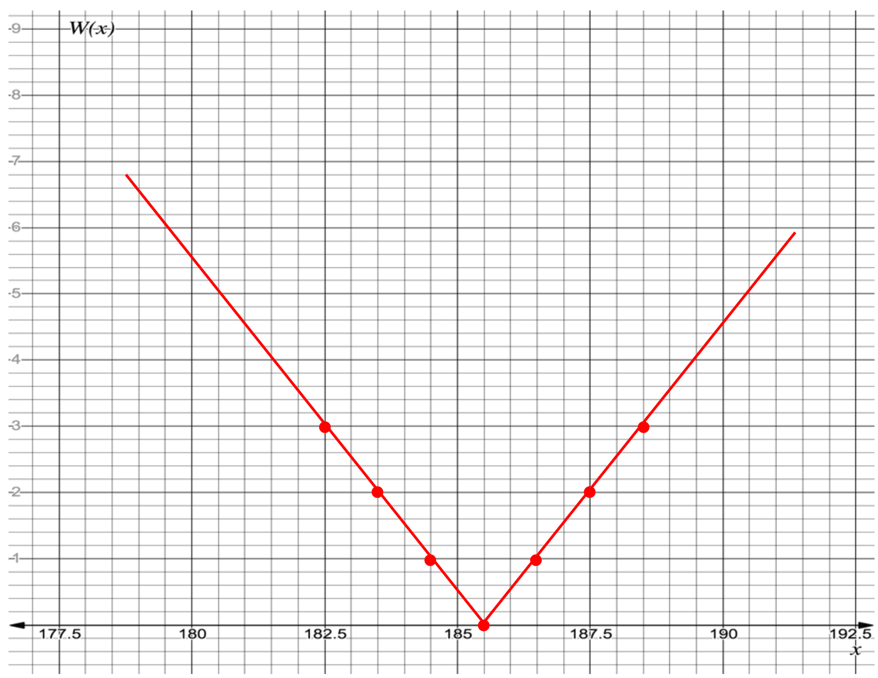
6. Determine the maximum and minimum clearance between the bridge and water.

minimum = 182.5 feet   maximum = 188.5 feet

7. Complete the table for $W(x) = |x - 185.5|$.

x	182.5	183.5	184.5	185.5	186.5	187.5	188.5
y	3	2	1	0	1	2	3

8. Sketch the graph of the absolute value function.



9. Interpret the vertex of $W(x) = |x - 185.5|$.

No fluctuation above or below the normal depth.

10. Interpret the axis of symmetry of $W(x) = |x - 185.5|$.

The line that determines how far away from the the normal depth, in feet, the water fluctuates.

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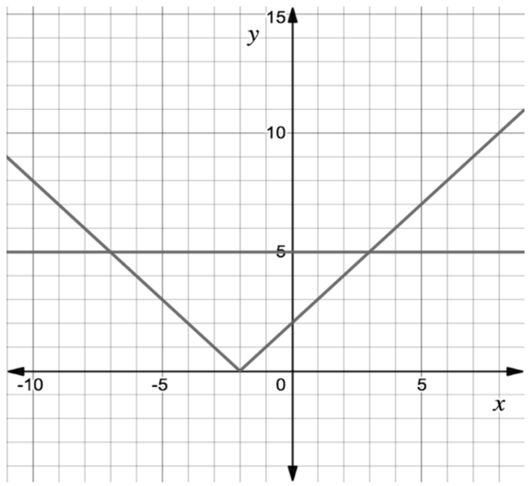
Lesson 3 Practice Period____________________

For questions 1-2, consider the absolute value equation $|x + 2| = 5$. The graphs of $y = |x + 2|$ and $y = 5$ are shown on the coordinate grid.



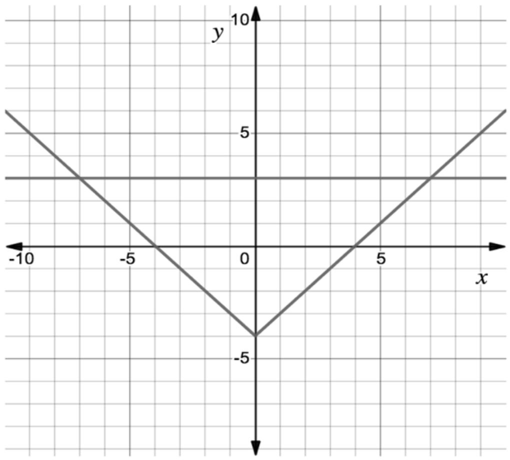
1. Use the graph to determine the solution to $|x + 2| = 5$.

   $x = 3$ and $x = -7$

2. Solve $|x + 2| = 5$ algebraically.

   $x + 2 = 5$ $x + 2 = -5$
   $x = 3$ $x = -7$

For questions 3-4, consider the absolute value equation $|x| - 4 = 3$. The graphs of $y = |x| - 4$ and $y = 3$ are shown on the coordinate grid.



3. Use the graph to determine the solution to $|x| - 4 = 3$.

$x = 7$ and $x = -7$

4. Solve $|x| - 4 = 3$ algebraically.

$|x| - 4 = 3$ $\quad$ $|x| = 7$

$x = 7$ $\quad$ $x = -7$

For questions 5-10, solve each absolute value equation.

5. $|m| - 37 = -12$

$+37 \quad +37$

$|m| = 25$

$m = 25 \quad m = -25$

6. $|h - 19| = 34$

$h - 19 = 34 \quad h - 19 = -34$

$h = 53 \quad h = -15$

7. $|4x| = 22$

$4x = 22 \quad 4x = -22$

$x = 5.5 \quad x = -5.5$

8. $21 = |x| + 9$

$12 = |x|$

$x = 12 \quad x = -12$

9. $|k - 49| = 68$

$k - 49 = 68 \quad k - 49 = -68$

$k = 117 \quad k = -19$

10. $21 = |9x|$

$9x = 21 \quad 9x = -21$

$x = 2.\overline{3} \quad x = -2.\overline{3}$

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Lesson 4 Practice Period______________________

Use the following context to answer questions 1-5.

The length of a machine part should be 6.75 inches (in). The manufacturing tolerance is within 0.04 in.

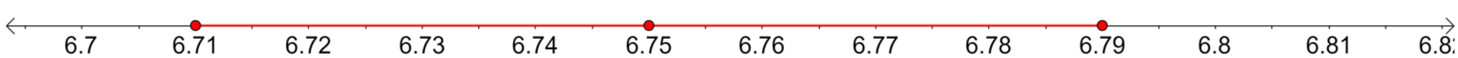
1. What is the minimum value the machine part could be?

$6.75 - 0.04 = 6.71$ in.

2. What is the maximum value the machine part could be?

$6.75 + 0.04 = 6.79$ in.

3. Sketch a number line that shows the minimum value, maximum value, and 6.75.



4. Write an absolute value equation that could be used to determine the maximum and minimum value of m, the length of the machine part, algebraically.

$|m - 6.75| = 0.04$

5. Show that the solutions to the equation match the maximum and minimum values determined in questions 1 and 2?

$$\begin{aligned} m - 6.75 &= 0.04 \\ +6.75 \quad &\;\; +6.75 \\ m &= 6.79 \end{aligned} \qquad \text{and} \qquad \begin{aligned} m - 6.75 &= -0.04 \\ +6.75 \quad &\;\; +6.75 \\ m &= 6.71 \end{aligned}$$

For questions 6-10, solve each absolute value equation.

6. $0 = 4|x+3| - 16$

$+16 \qquad +16$

$\frac{16}{4} = \frac{4|x+3|}{4}$

$4 = |x+3|$

$x + 3 = 4$ and $x + 3 = -4$

$-3 \quad -3 \qquad\qquad -3 \quad -3$

$x = 1 \qquad\qquad x = -7$

$x = 1$ and -7

7. $|-2x+7| - 21 = 0$

$+21 \quad +21$

$|-2x+7| = 21$

$-2x + 7 = 21$ and $-2x + 7 = -21$

$-7 \quad -7 \qquad\qquad -7 \quad -7$

$\frac{-2x}{-2} = \frac{14}{-2} \qquad\qquad \frac{-2x}{-2} = \frac{-28}{-2}$

$x = -7 \qquad\qquad x = 14$

-7 and 14

8. $|x+3| + 17 = 11$

$-17 \quad -17$

$|x+3| = -6$

No solution

9. $|4x+5| - 8 = 29$

$+8 \quad +8$

$|4x+5| = 37$

$4x + 5 = 37$ and $4x + 5 = -37$

$-5 \quad -5 \qquad\qquad -5 \quad -5$

$\frac{4x}{4} = \frac{32}{4} \qquad\qquad \frac{4x}{4} = \frac{-42}{4}$

$x = 8 \qquad\qquad x = -10.5$

-10.5 and 8

10. $-|x-2| + 9 = 0$

$-9 \quad -9$

$\frac{-|x-2|}{-1} = \frac{-9}{-1}$

$|x-2| = 9$

$x - 2 = 9 \qquad x - 2 = -9$

$+2 \quad +2 \qquad +2 \quad +2$

$x = 11 \qquad x = -7$

-7 and 11

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For questions 1-2, consider the absolute value inequality $|5x - 8| - 10 < 17$.

1. Solve the inequality.

$|5x - 8| \cancel{-10} < 17$

$\underline{\quad\quad \cancel{+10} + 10}$

$|5x - 8| < 27$

$\boxed{-3\frac{4}{5} < x < 7}$

$5x \cancel{-8} > -27$

$\underline{\quad \cancel{+8} \quad +8}$

$\frac{5x}{5} > \frac{-19}{5}$

$\boxed{x > -3\frac{4}{5}}$

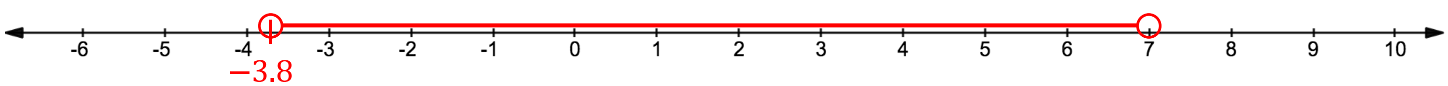
$5x \cancel{-8} < 27$

$\underline{\quad \cancel{+8} \quad +8}$

$\frac{5x}{5} < \frac{35}{5}$

$\boxed{x < 7}$

2. Graph the solution on the number line.



For questions 3-4, consider the absolute value inequality $-6|x - 7| - 12 < 0$.

3. Solve the inequality.

$-6|x - 7| \cancel{-12} < 0$

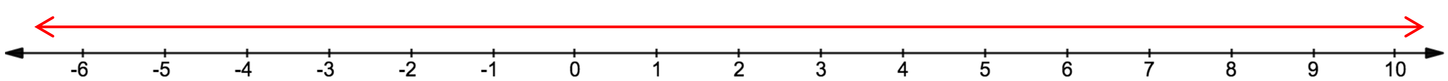
$\underline{\quad\quad \cancel{+12} + 12}$

$\frac{-6|x - 7|}{-6} < \frac{12}{-6}$

$|x - 7| > -2$

infinite solutions

4. Graph the solution on the number line.



For questions 5-6, consider the absolute value inequality $|4x - 8| \geq 16$.

5. Solve the inequality.

$|4x - 8| \geq 16$

$4x - 8 \leq -16$

$\underline{\quad +8 \quad\quad +8}$

$\frac{4x}{4} \leq \frac{-8}{4}$

$\boxed{x \leq -2}$

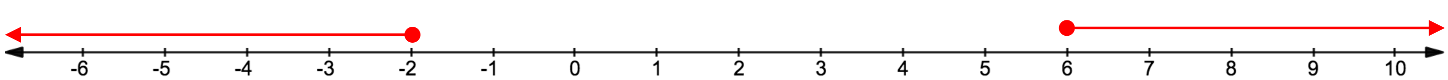
$4x - 8 \geq 16$

$\underline{\quad +8 \quad +8}$

$\frac{4x}{4} \geq \frac{24}{4}$

$\boxed{x \geq 6}$

6. Graph the solution on the number line.



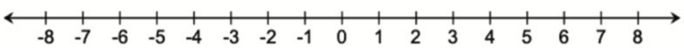
For questions 7-10, solve the absolute value inequality. The graph the solution on the number line.

7. $\cancel{-7}|x-11| > 28$

$\underline{\cancel{-7}} \quad \underline{-7}$

$|x-11| < -4$

No Solution



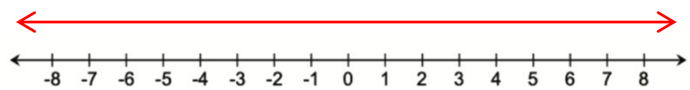
8. $-4|x-5| - \cancel{8} \leq 28$

$\cancel{+8} \quad +8$

$\frac{-4|x-5|}{-4} \leq \frac{36}{-4}$

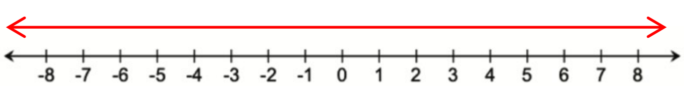
$|x-5| \geq -9$

Infinite solutions



9. $|x-17| \geq -5$

Infinite solutions



10. $|4x+5| - \cancel{9} < 12$

$\underline{\cancel{+9} \quad +9}$

$|4x+5| < 21$

$4x + \cancel{5} > -21$

$\underline{\cancel{-5} \quad -5}$

$\frac{4x}{4} > \frac{-26}{4}$

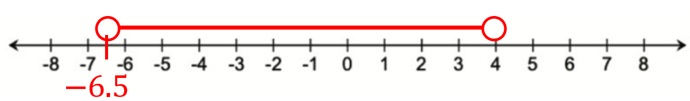
$x > -6\frac{1}{2}$

$4x + \cancel{5} < 21$

$\underline{\cancel{-5} \quad -5}$

$\frac{4x}{4} < \frac{16}{4}$

$x < 4$



$-6\frac{1}{2} < x < 4$

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Use the following context to answer questions 1-4.

Members of the cross country team are preparing for a 3-mile race. The average time for the team is 21.25 minutes. The fastest and slowest times varied from the average by 3.1 minutes.

1. Define the variable.

   Let $x =$ number of minutes

2. Write an absolute value inequality that represents the context.

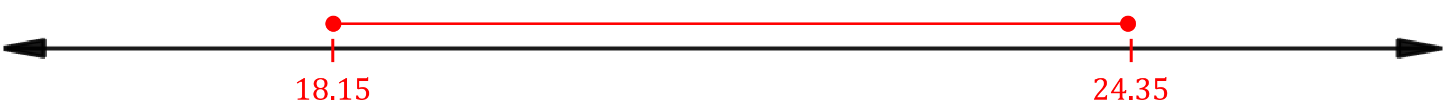
   $|x - 21.25| \leq 3.1$

3. Solve the inequality. Write the answer as a compound inequality.

$$\begin{array}{rr} x - \cancel{21.25} \geq -3.1 \\ \underline{+\cancel{21.25} \quad +21.25} \\ x \geq 18.15 \end{array} \qquad \begin{array}{rr} x - \cancel{21.25} \leq 3.1 \\ \underline{+\cancel{21.25} \quad +21.25} \\ x \leq 24.35 \end{array}$$

$$18.15 \leq x \leq 24.35$$

4. Graph the solution on the number line.



Use the following context to answer questions 5-6.

Noah is a member of the high school band. Each day he spends an average of 62 minutes practicing for an upcoming band concert. He spent less time practicing on school days, but more time on the weekends. Each of his daily practice times was within 29 minutes of his average.

5. Write an absolute value inequality that represents m, the number of minutes Noah spends practicing for the band concert on a given day.

   $|m-62| \leq 29$

6. Solve the inequality.

   $$\begin{aligned} -29 &\leq m-62 \leq 29 \\ \underline{+62} & \quad \underline{+62 \quad +62} \\ 33 &\leq m \leq 91 \end{aligned}$$

   between 33 & 91 minutes

Use the following context to answer questions 7-10.

Payton made a personal goal of earning $\$145$ per week at her after-school job. Each week, she has been averaging within $\$19.23$ of her goal.

7. Define the variable.

   Let $x =$ the amount of her weekly paycheck.

8. Write an absolute value inequality that represents the context.

   $|x-145| \leq 19.23$

9. Solve the inequality. Write the answer using set-builder notation.

   $$\begin{aligned} -19.23 &\leq x-145 \leq 19.23 \\ \underline{+145} & \quad \underline{+145 \quad +145} \\ 125.77 &\leq x \leq 164.23 \end{aligned}$$

   $\{x \mid 125.77 \leq x \leq 164.23\}$

10. Graph the solution on the number line.